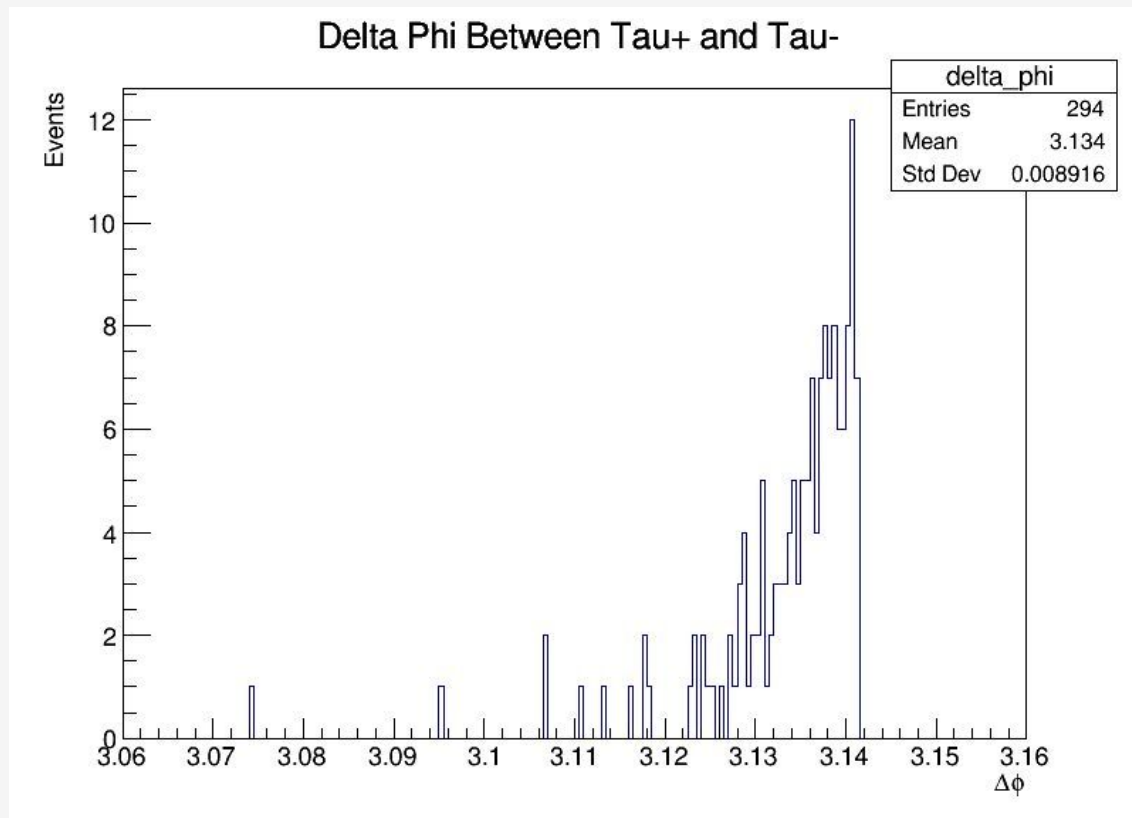


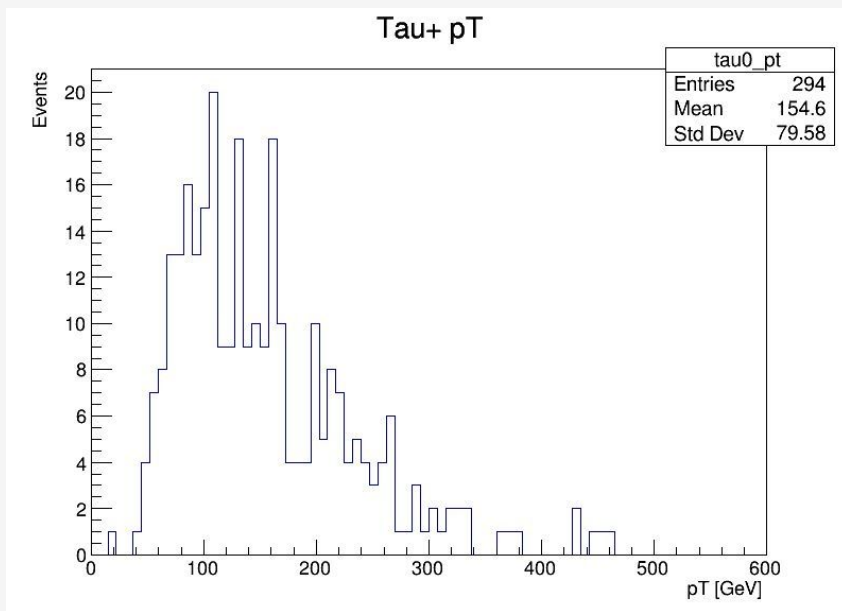
01

Variables and distributions

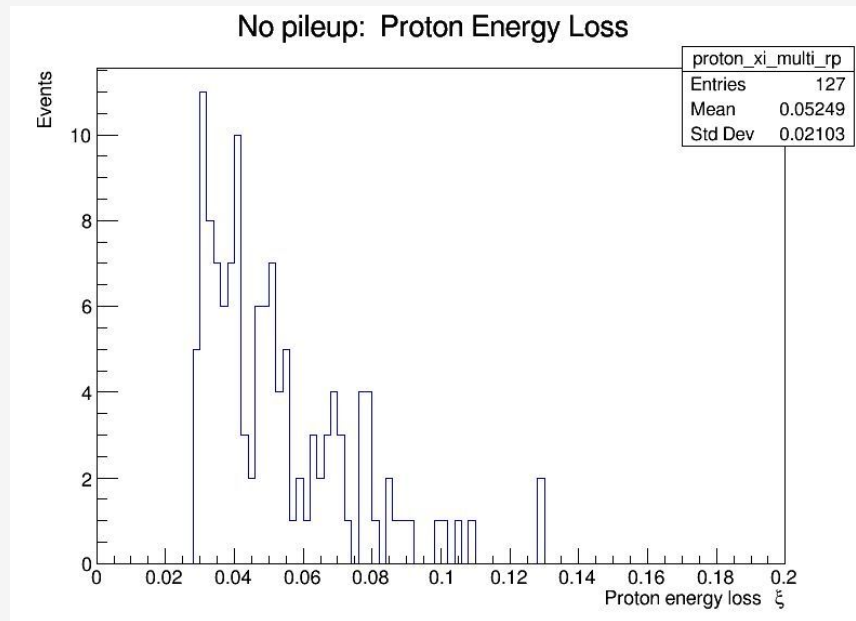
$\Delta\phi$



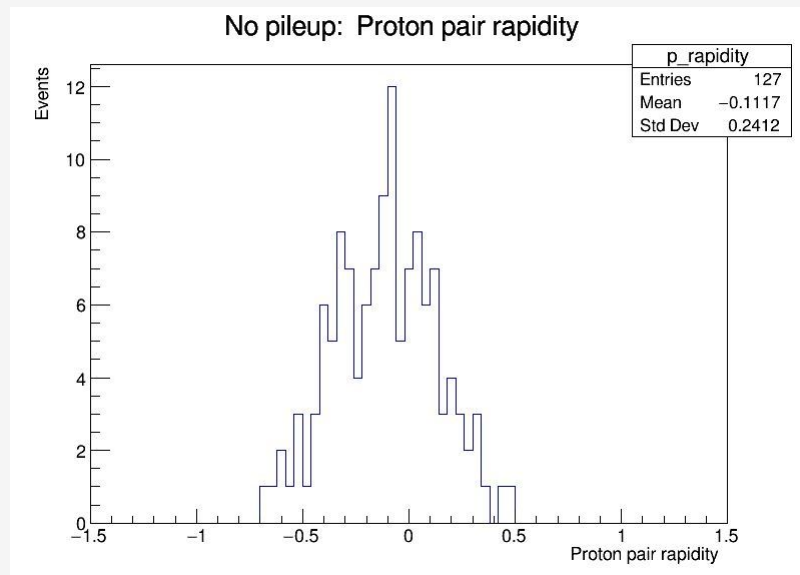
Pseudorapidity, η



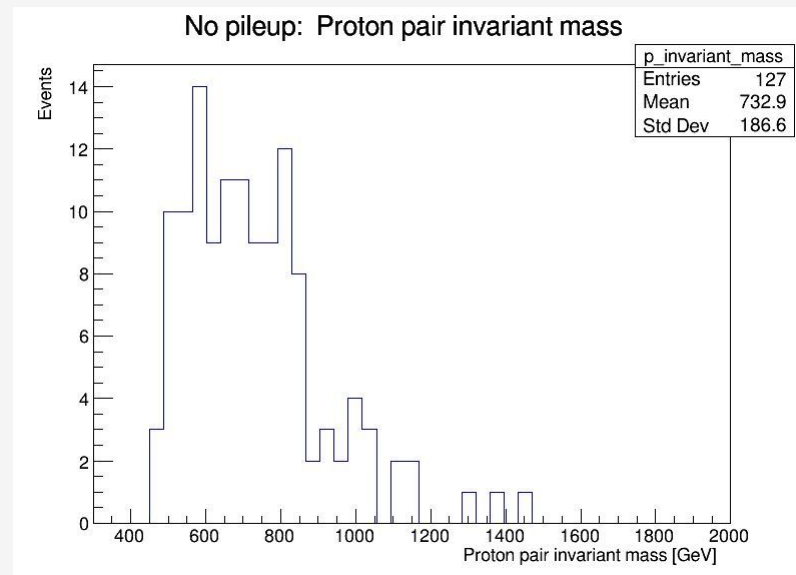
Proton energy Loss, ξ



Proton pair rapidity

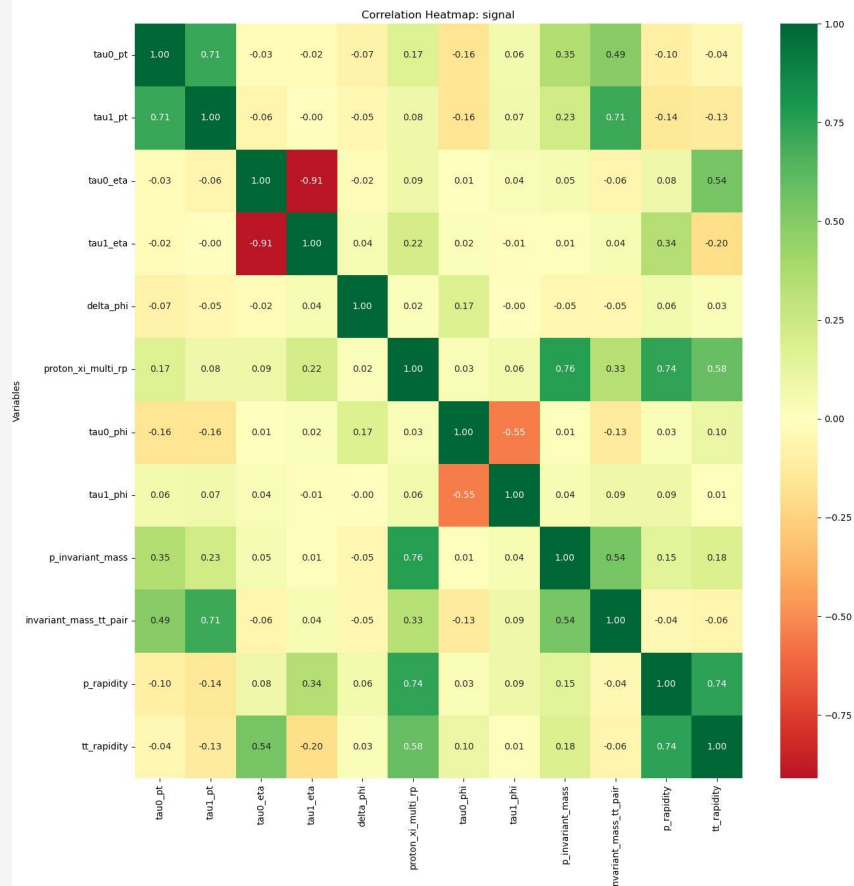


Proton pair invariant mass



02

Correlation between variables

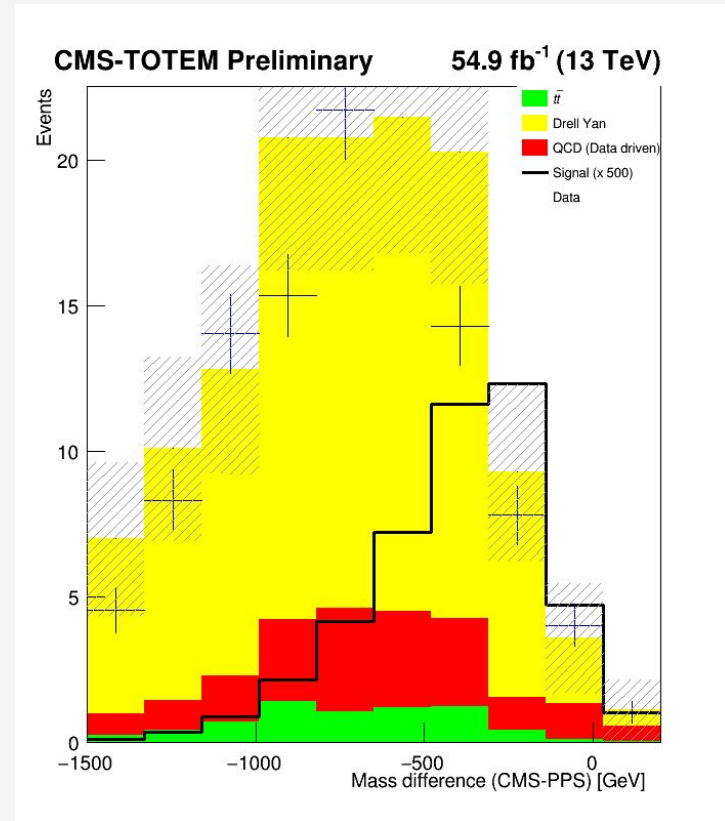
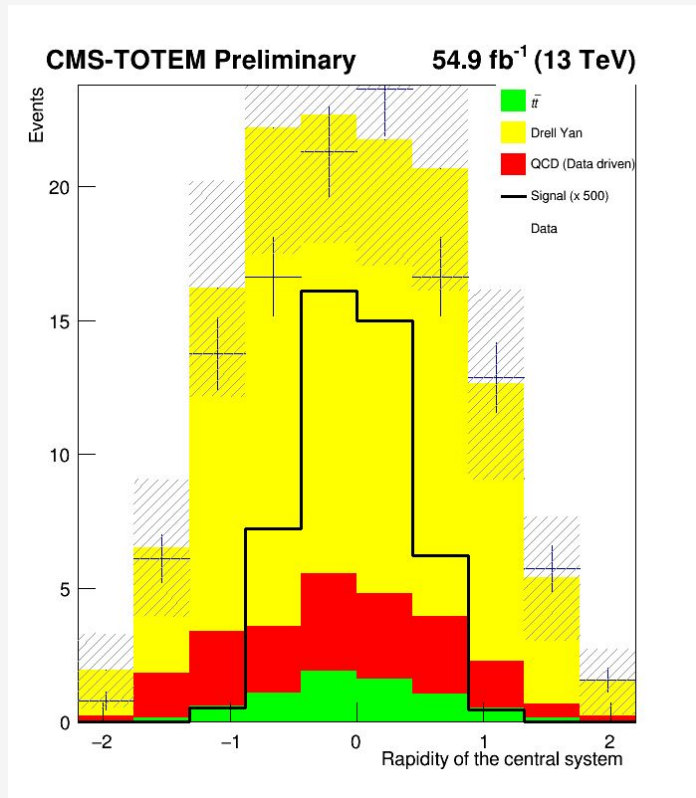


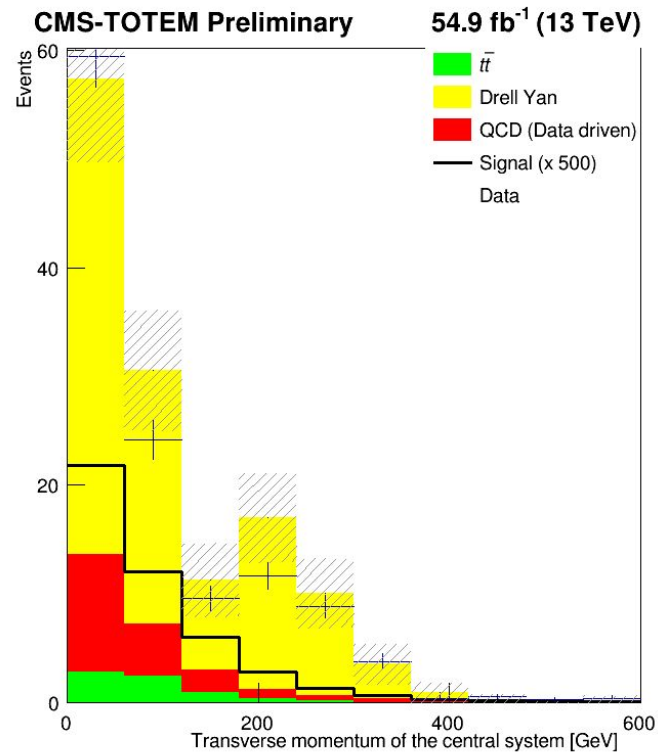
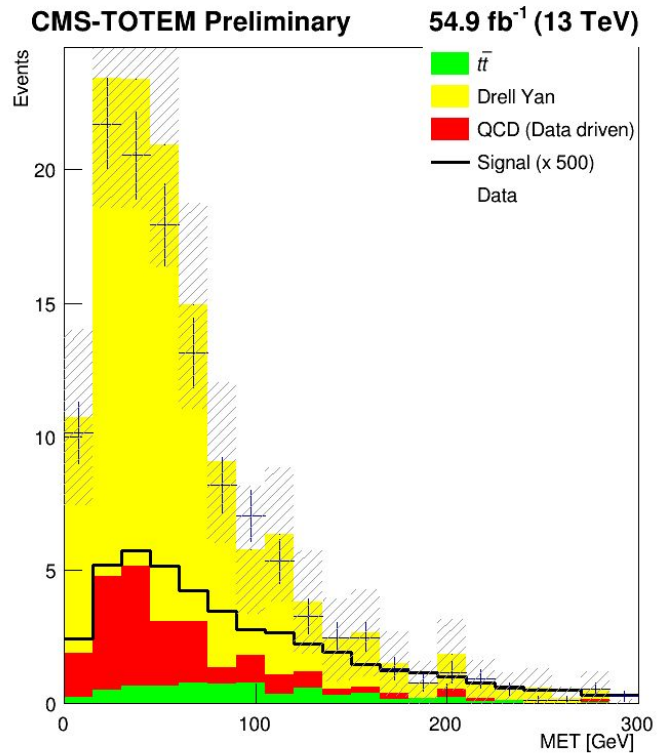
Expected:

- Strong correlation between tau0 and tau1 phi
- Negative correlation between tau0 and tau1 eta
- Strong correlation between invariant mass of tau pair and tau0 and tau1 pt
- Some correlation between tau pair rapidity and tau0 eta

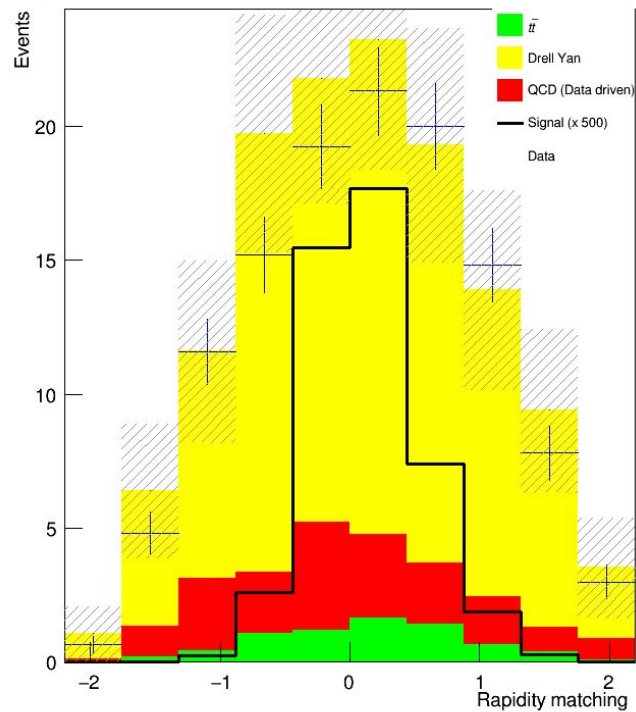
Input distributions to the BDT discriminator

- Invariant mass of the central system
→ The total mass of all particles in the central detector, calculated from their 4-momenta.
- Rapidity of the central system
→ Describes how forward or backward the central system is moving along the beam axis.
- Mass difference
→ Difference between the central system mass and the mass reconstructed using the forward protons (CMS - PPS).
- MET (Missing Transverse Energy)
→ Represents undetected particles.
- Transverse momentum of the central system
→ Momentum of the central system perpendicular to the beam axis.
- Rapidity matching
→ Comparison of rapidity measured centrally..
- Transverse momentum of hadronic tau
→ Momentum perpendicular to the beam of the hadronically decaying tau lepton.
- Acoplanarity of central system
→ Measures how "non-back-to-back" the two leading tau candidates are; zero means perfect balance.

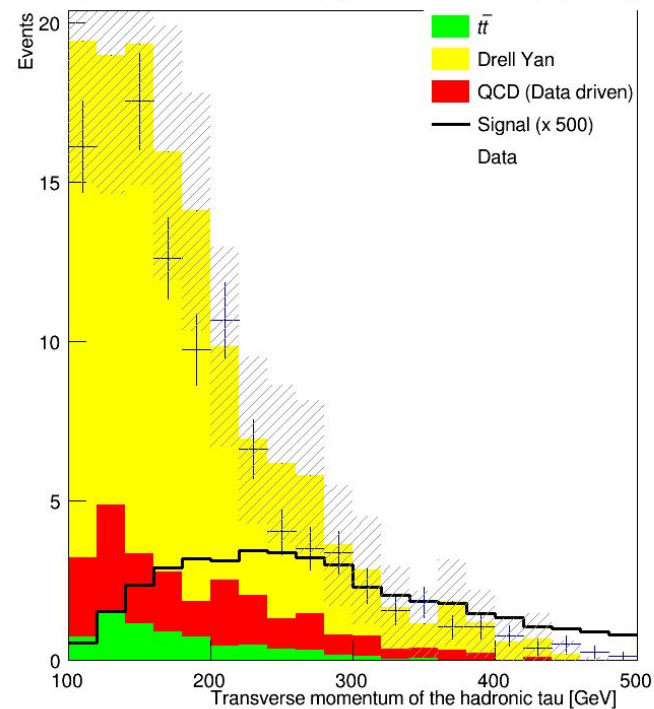




CMS-TOTEM Preliminary **54.9 fb⁻¹ (13 TeV)**

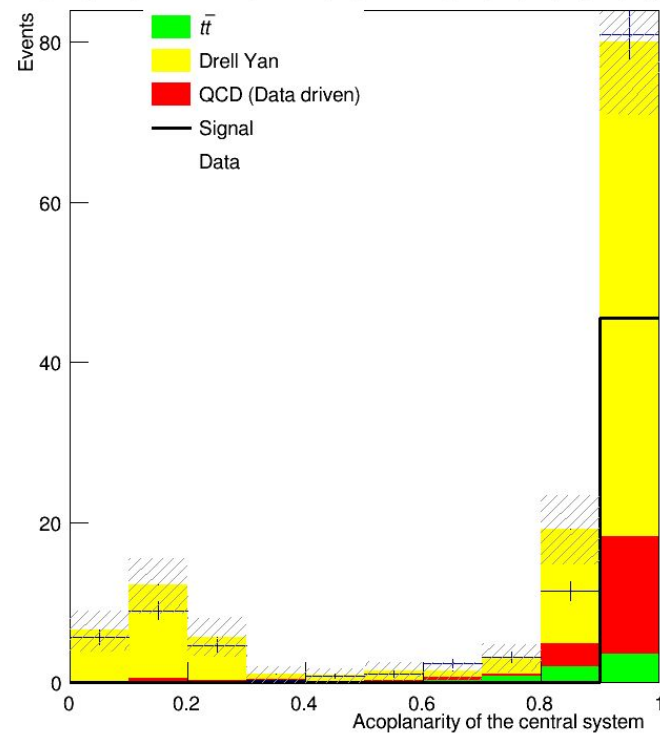


CMS-TOTEM Preliminary **54.9 fb⁻¹ (13 TeV)**



CMS-TOTEM Preliminary

54.9 fb⁻¹ (13 TeV)



Weights applied in the previous plots

189 In 2018, due to an improved PPS detector configuration, silicon pixels are used. They allow the
190 reconstruction of multiple protons and the scenario is different. Therefore, the technique used
191 is the following:

- 192 • We estimate the probability $p_{n,m}$ to get n protons in arm 0 and m protons in arm 1;
- 193 • For the background samples, we observed that the probability to have at least one
194 PU proton per arm is just 13%. Therefore, the weight of each background event is
195 rescaled of a factor 0.13;
- 196 • Subsequently, each background event is enriched with n protons in arm 0 and m
197 protons in arm 1, according to the probability $p_{n,m}$ calculated earlier. If we get more
198 than one proton per arm, the pair with the largest ζ is chosen;
- 199 • For the signal samples, they are directly enriched with the number of protons calcu-
200 lated according to the probability $p_{n,m}$.

```
//double w_data=1.;  
double w_qcd=1.*0.8*0.13;  
double w_ttjets=0.15*0.8*0.13;  
double w_dy=1.81*0.8*0.13;  
double w_sinal;  
double w_data=1*0.13;
```

```
double weight = .15; //w= Numero monte carlo / Numero de  
2018 , Numero de 2018 = luminosidade de 2018 * secção de  
2018
```

0.13 is due to pps reconstruction technique -> probability of having at least one proton per arm

Still no explanation for 0.8 factor on the background only
-> maybe some arbitrary normalization factor

03

Code specifications

Dataflow

Background

Starts with data in NANOAOD format
DY and ttbar are MC
QCD is based on same sign events from data

Fase0
↓
QCD_fase0.cpp
DY_fase0.cpp
Ttjets_fase0.cpp

↓
Produces fase0 files
for each background
used in:

Fase1
↓
nanotry_fase1_QCD.cpp
nanotry_fase1_DY.cpp
nanotry_fase1_jets.cpp

↓
Produces fase1, one
file for each, after
enriched with pileup
protons

→ QCD_2018_UL_skimmed_TauTau_nano_faseltotal-protons_2018.root
→ DY_2018_UL_skimmed_TauTau_nano_faseltotal-protons_2018.root
→ ttJets_2018_UL_skimmed_TauTau_nano_faseltotal-protons_2018.root

final data files with the signal-like selection (opposite sign taus) :
Dados_2018_UL_skimmed_TauTau_nano_faseltotal-
protons_2018.root

Signal

Starts with data in MINIAOD format

↓
Sinal.cpp

→ data/MC corrections for the efficiency,
scale, and resolution of taus, electrons,
muons, and protons

↓
Produces ntuple with format similar as fase1,
includes BSM weighs needed to normalize the
sample for different SM and BSM scenarios

↓
TauTau_sinal_PIC_july_2018.root

Used in plot_m.cpp
Produces normalized plots and the input distributions for
BDT

Fase0: general description

- Tau ID variables (DeepTau),
- Tau momenta (pt, eta, phi, mass),
System variables (mass, pt, acoplanarity, rapidity),
- Jet info (pt, b-tagging),
- MET (missing transverse energy),
Generator-level info.

Calculates:

- Invariant mass of the tau pair,
- Acoplanarity between taus,
- System pt and rapidity,
- Number of b-tagged jets (DeepB > 0.4506),

Applies trigger filter (`HLT_DoubleMediumChargedIsoPFTauHPS40_Trk1_eta2p1_Reg == 1`)

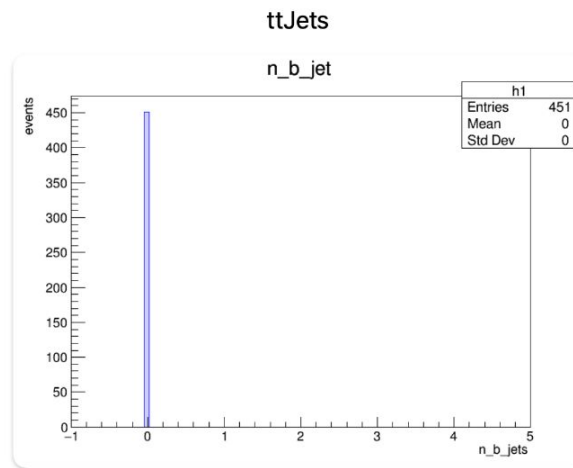
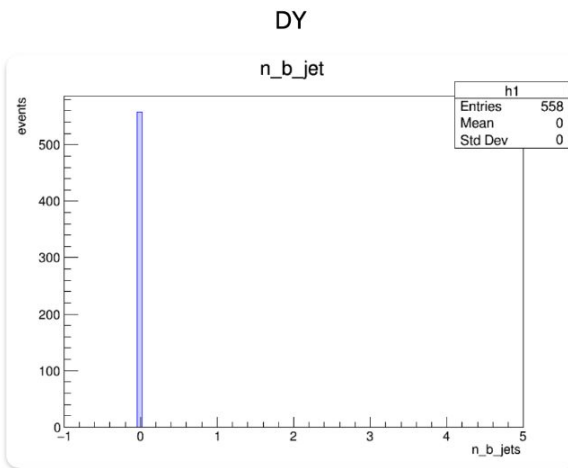
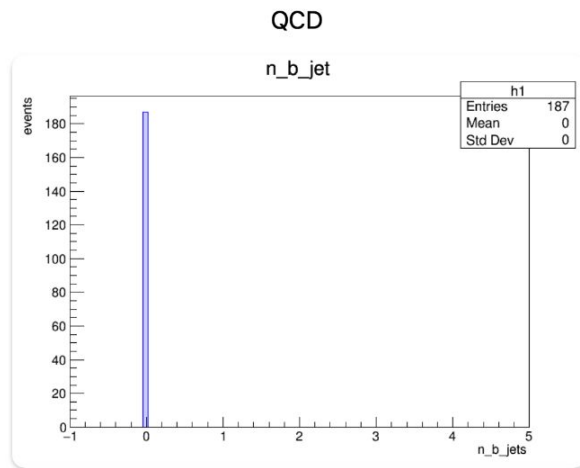
Code organization Background

Goal: only one file for signal and one file for background
(with parallelization using `rootEnableImplicit`)

File name	reads		Output
DY_fase0	DY 2018 UL.txt		DY 2018 UL skimmed Tau Tau nano
QCD_fase0	QCD 2018 UL.txt dadosluminosidade.txt	Checks if luminosity in event matches with the data from file	QCD 2018 UL skimmed Ta uTau nano
ttJets_fase0	ttJets 2018 UL.txt		ttJets 2018 UL skimmed TauTau nano
nanotry_fase1_DY	DY 2018.txt Name of fase0 output files		
nanotry_fase1_QCD	QCD fase1 PIC.txt Name of fase0 output files		
nanotry_fase1_ttJets	ttJets fase1 PIC.txt Name of fase0 output files		

Number of b jets: 0 for all samples

Fase1

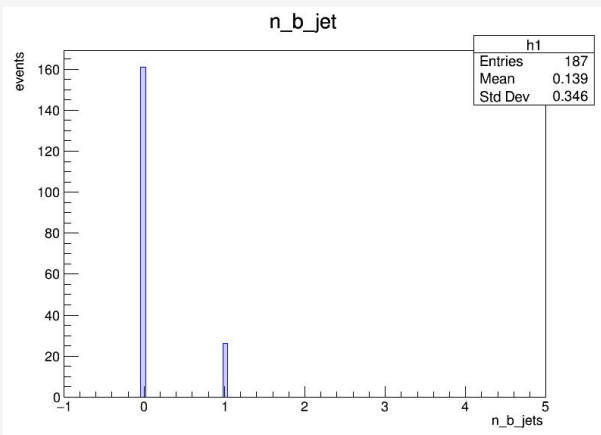


Cause: mismatch in the declaration of variable type in the proton enrichment code

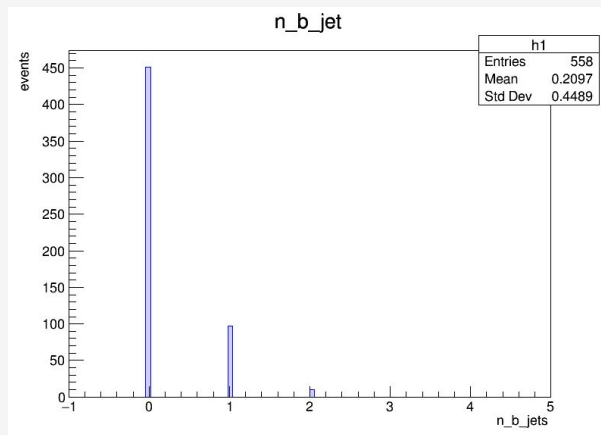
- Mismatch in the declaration of variable in the proton enrichment code (to introduce pileup protons);
 - N_b_jet was declared as *double* when it originally was *int*
- Handling of branch addresses

-> Fixed by passing the merged samples from phase1 through the fixed proton enrichment code

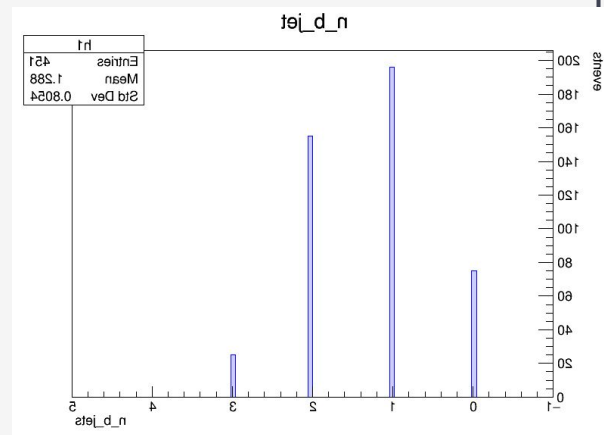
After fixing n_b_jet: Fase 1 samples



qcd



dy



ttjets

Control region observations

- worse data/simulation ratio for higher values on most variables;
- In (original) QCD CR the range of the plot of invariant mass of the central system [400,1000] does not add up with the limits of the control region [100,300] -> this was changed in the new plots;
- Also in QCD CR there are some discrepancies between the shape of the invariant mass of the central system and transverse momentum of central system plots;
- In TTJETS CR the cutoff of the transverse momentum also doesn't match the CR cut (in the original the plot goes to around 240 GeV, but the cut is $p_t < 125$ GeV);
- In the DY CR plots, the shapes look mostly similar, except for the invariant mass of the central system.

Exploring new physics with proton interactions at the LHC

Process: $pp \rightarrow pX(\rightarrow \tau\tau)p$;

Protons are tagged by PPS, while the decay products of the X system are detected by the CMS central detector.

Decay mode in study:

- Fully hadronic decay mode; Both τ s decay hadronically

Index

Slide 4: Variables and distributions

Slide 8: Correlation between variables

Slide 17: Code specifications

Slide 23: BDT training: specifications

Slide 26: Control regions

Slide 32: Control region observations

Weights applied in the previous plots

In original code

```
weight_sample = 54900.*(1.)/(4000.*1000.);
```

- Weight_sample is normalization factor
- weight_sample=0.013725

$$\text{weight_sample} = \frac{\mathcal{L} \cdot \sigma}{N_{\text{gen}}}$$

What worked to make the plots the same:

Is original weight_sample/1000 -> UNITS

```
weight_sample= 0.000013725;
```

Code organization
Background-> fase0 specifics

Feature	DY	ttjets	QCD
Sample Type	Drell-Yan MC	t $\bar{t}$ MC	Data-driven QCD
Luminosity block filtering			✓
Extra gen matching	✓	✓	
Generator weight	1.81	0.15	
HLT version	40_Trk1_eta2p1	40_Trk1_eta2p1	35_Trk1_eta2p1

Code organization
Background-> fase1 specifics

Feature	DY	ttjets	QCD
Sample Type	Drell-Yan MC	t $\bar{t}$ MC	Data-driven QCD
Scale Factors	Tau ID + energy scale + sys	Tau ID + energy scale	
Systematic Variations	full		
Charge Selection	Opposite Sign (<0)	Opposite Sign (<0)	Same Sign (>0)
Tau Selection Cuts	$p_t > 100, \eta < 2.3$	$p_t > 100, \eta < 2.4$	$p_t > 100, \eta < 2.4$
Gen-Match Handling	yes		

BDT training: specifications

Variables used:

Variable	Description	Units
<code>sist_acop</code>	Central system acoplanarity	dimensionless
<code>sist_pt</code>	Total momentum	GeV
<code>sist_mass</code>	Invariant mass	GeV
<code>sist_rap</code>	System rapidity	dimensionless
<code>met_pt</code>	Missing energy	GeV
Mass matching	<code>sist_mass - sqrt(13000² × $\xi_1 \times \xi_2$)</code>	GeV
Rapidity matching	<code>sist_rap - 0.5 × log(ξ_1 / ξ_2)</code>	dimensionless

TMVAClassification	Train BDT/ML models	Background file Signal file	Output_TMVA_tautau_2018.root Dataset weights
TMVAClassificationApplication	Apply trained models to data	Trained models Data samples: dy, ttjets,qcd and signal	On file for each type containing BDT scores for every event
Bdt_output	Process BDT results	TMVApp_*.root Event weights, sample information	Analysis histograms Bdt score distributions signal/background separation cuts Cut optimization results
TMVAOutPut_Distributions	Create TMVA evaluation plots	Output_TMVA_tautau_2018.root Training results Test results Method performance data	ROC curves Variable importance plots Correlation matrices

04

Control regions

In red: not applied

In green: applied in phase 0 or 1

Control regions: $t\bar{t}h$

	Invariant mass (GeV)	acoplanarity	η	2 isolated ts	N of b-jets	p_t
DY	[40,100]	<0.3	<2.4	Opposite sign	0	
QCD	[100,300]	>0.8	<2.4	Same sign	0	<75
ttjets	[200,650]	>0.5	<2.4	Opposite sign	1 or more	<125

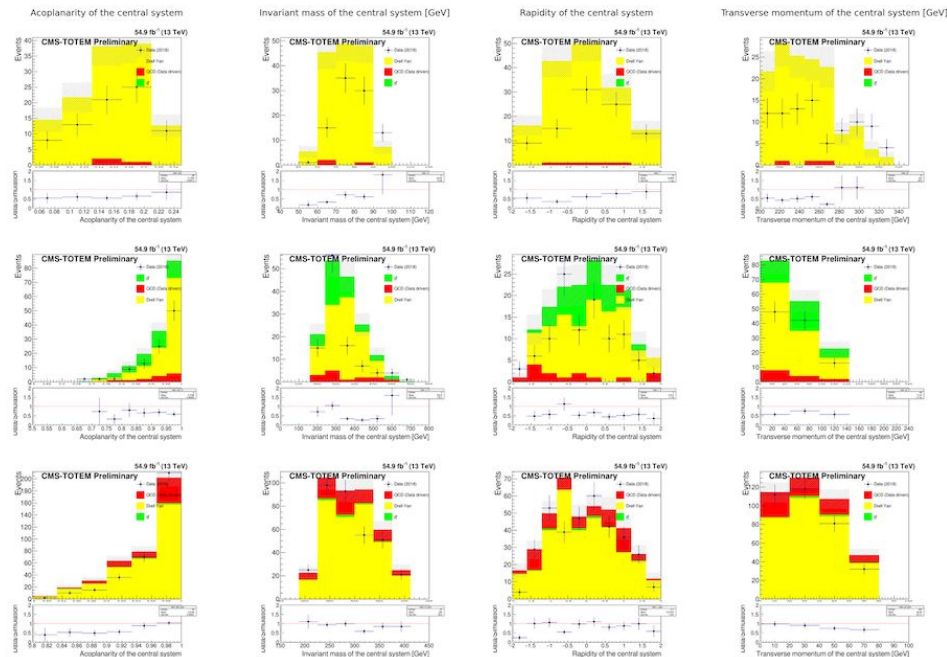
On top: original from analysis note

Bottom: new plots

TT CR



32



samples used and information

Background	Dataset	Cross section (pb)
Drell Yan (DY)	/DYJetsToLL_M*_TuneCP5_13TeV-amcatnloFXFX-pythia8/RunIISummer20UL18NanoAODv9-106X*/NANOAODSIM	6404
inclusive $t\bar{t}$	TTJets_TuneCP5_13TeV-amcatnloFXFX-pythia8/NANOAODSIM/106X_upgrade2018_realistic_v16_L1v1-v1	832
QCD, $e\tau_h$ channel	Run2018*/EGamma/NANOAOD/UL2018_MiniAODv2_NanoAODv9-v1	data-driven – luminosity 54.9 fb^{-1}
QCD, $\mu\tau_h$ channel	Run2018*/SingleMuon/NANOAOD/UL2018_MiniAODv2_NanoAODv9-v2	data-driven – luminosity 54.9 fb^{-1}

Table 3: List of the background considered. As for the Drell-Yan (DY), only events with M_X in the range $[50, 3000]$ GeV are considered (according to PPS acceptance [6]). For QCD, a data-driven approach is considered and data are used. This table refers to the 2018 data-taking period.

Questions

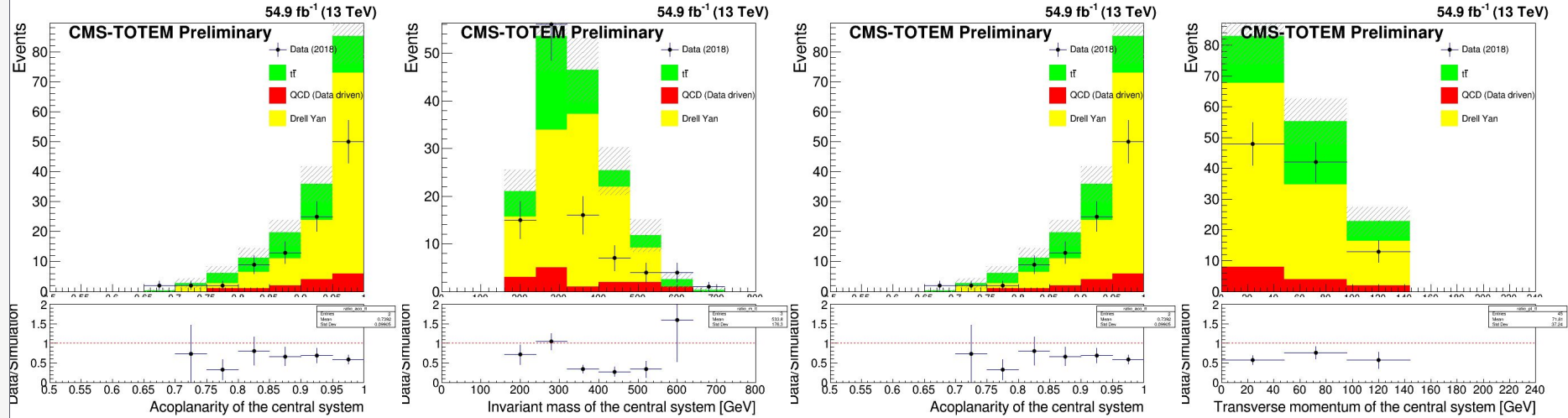
- The samples I am using (the ones mentioned in the twiki) come from the same datasets (referenced in the previous slide)?
-

tau mass [40,100]
system acoplanarity <0.3
n° of b jets =0



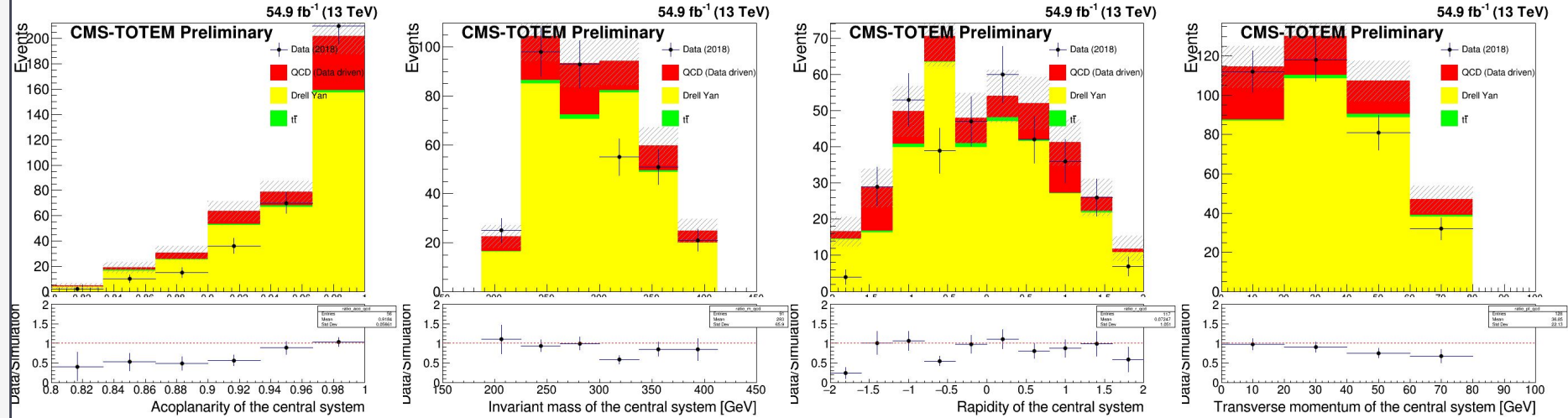
TT Control region

tau mass [200,650]
 system acoplanarity > 0.5
 n° of b jets > 0
 tau pair pt < 125



QCD control region

tau mass [100,400]
 system acoplanarity > 0.8
 n° of b jets = 0
 tau pair pt < 75



title

title

title

title

To dos:

In progress:

- Writing code to parallelize phase0 for all samples (using Rdataframe):
 - DY
 - QCD
 - TTjets ✓ (from 2-3 days to ~ 2 hours)
 - data
 - signal